

Modeling with First-Order Differential Equations



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- 3.1** Linear Models
- 3.2** Nonlinear Models
- 3.3** Modeling with Systems of First-Order DEs

CHAPTER 3 IN REVIEW

In Section 1.3 we saw how a first-order differential equation could be used as a mathematical model in the study of population growth, radioactive decay, cooling of bodies, mixtures, chemical reactions, fluid draining from a tank, velocity of a falling body, and current in a series circuit. Using the methods discussed in Chapter 2, we are now able to solve some of the linear DEs in Section 3.1 and nonlinear DEs in Section 3.2 that frequently appear in applications.

3.1 Linear Models

INTRODUCTION In this section we solve some of the linear first-order models that were introduced in Section 1.3.

GROWTH AND DECAY The initial-value problem

$$\frac{dx}{dt} = kx, \quad x(t_0) = x_0, \quad (1)$$

where k is a constant of proportionality, serves as a model for diverse phenomena involving either growth or decay. We saw in Section 1.3 that in biological applications the rate of growth of certain populations (bacteria, small animals) over short periods of time is proportional to the population present at time t . Knowing the population at some arbitrary initial time t_0 , we can then use the solution of (1) to predict the population in the future—that is, at times $t > t_0$. The constant of proportionality k in (1) can be determined from the solution of the initial-value problem, using a subsequent measurement of x at a time $t_1 > t_0$. In physics and chemistry (1) is seen in the form of a *first-order reaction*—that is, a reaction whose rate, or velocity, dx/dt is directly proportional to the amount x of a substance that is unconverted or remaining at time t . The decomposition, or decay, of U-238 (uranium) by radioactivity into Th-234 (thorium) is a first-order reaction.

EXAMPLE 1 Bacterial Growth

A culture initially has P_0 number of bacteria. At $t = 1$ h the number of bacteria is measured to be $\frac{3}{2}P_0$. If the rate of growth is proportional to the number of bacteria $P(t)$ present at time t , determine the time necessary for the number of bacteria to triple.

SOLUTION We first solve the differential equation in (1), with the symbol x replaced by P . With $t_0 = 0$ the initial condition is $P(0) = P_0$. We then use the empirical observation that $P(1) = \frac{3}{2}P_0$ to determine the constant of proportionality k .

Notice that the differential equation $dP/dt = kP$ is both separable and linear. When it is put in the standard form of a linear first-order DE,

$$\frac{dP}{dt} - kP = 0,$$

we can see by inspection that the integrating factor is e^{-kt} . Multiplying both sides of the equation by this term and integrating gives, in turn,

$$\frac{d}{dt} [e^{-kt}P] = 0 \quad \text{and} \quad e^{-kt}P = c.$$

Therefore $P(t) = ce^{kt}$. At $t = 0$ it follows that $P_0 = ce^0 = c$, so $P(t) = P_0e^{kt}$. At $t = 1$ we have $\frac{3}{2}P_0 = P_0e^k$ or $e^k = \frac{3}{2}$. From the last equation we get $k = \ln \frac{3}{2} = 0.4055$, so $P(t) = P_0e^{0.4055t}$. To find the time at which the number of bacteria has tripled, we solve $3P_0 = P_0e^{0.4055t}$ for t . It follows that $0.4055t = \ln 3$, or

$$t = \frac{\ln 3}{0.4055} \approx 2.71 \text{ h.}$$

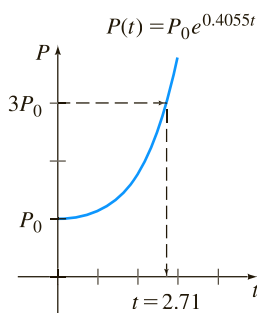


FIGURE 3.1.1 Time in which population triples in Example 1

See Figure 3.1.1.

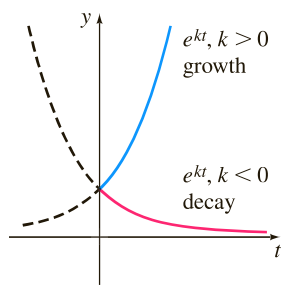


FIGURE 3.1.2 Growth ($k > 0$) and decay ($k < 0$)

Notice in Example 1 that the actual number P_0 of bacteria present at time $t = 0$ played no part in determining the time required for the number in the culture to triple. The time necessary for an initial population of, say, 100 or 1,000,000 bacteria to triple is still approximately 2.71 hours.

As shown in Figure 3.1.2, the exponential function e^{kt} increases as t increases for $k > 0$ and decreases as t increases for $k < 0$. Thus problems describing growth (whether of populations, bacteria, or even capital) are characterized by a positive value of k , whereas problems involving decay (as in radioactive disintegration) yield a negative k value. Accordingly, we say that k is either a **growth constant** ($k > 0$) or a **decay constant** ($k < 0$).

HALF-LIFE In physics the **half-life** is a measure of the stability of a radioactive substance. The half-life is simply the time it takes for one-half of the atoms in an initial amount A_0 to disintegrate, or transmute, into the atoms of another element. The longer the half-life of a substance, the more stable it is. For example, the half-life of highly radioactive radium, Ra-226, is about 1700 years. In 1700 years one-half of a given quantity of Ra-226 is transmuted into radon, Rn-222. The most commonly occurring uranium isotope, U-238, has a half-life of approximately 4,500,000,000 years. In about 4.5 billion years, one-half of a quantity of U-238 is transmuted into lead, Pb-206.

EXAMPLE 2 Half-Life of Plutonium

A breeder reactor converts relatively stable uranium-238 into the isotope plutonium-239. After 15 years it is determined that 0.043% of the initial amount A_0 of plutonium has disintegrated. Find the half-life of this isotope if the rate of disintegration is proportional to the amount remaining.

SOLUTION Let $A(t)$ denote the amount of plutonium remaining at time t . As in Example 1 the solution of the initial-value problem

$$\frac{dA}{dt} = kA, \quad A(0) = A_0$$

is $A(t) = A_0 e^{kt}$. If 0.043% of the atoms of A_0 have disintegrated, then 99.957% of the substance remains. To find the decay constant k , we use $0.99957A_0 = A(15)$ —that is, $0.99957A_0 = A_0 e^{15k}$. Solving for k then gives $k = \frac{1}{15} \ln 0.99957 = -0.00002867$. Hence $A(t) = A_0 e^{-0.00002867t}$. Now the half-life is the corresponding value of time at which $A(t) = \frac{1}{2}A_0$. Solving for t gives $\frac{1}{2}A_0 = A_0 e^{-0.00002867t}$, or $\frac{1}{2} = e^{-0.00002867t}$. The last equation yields

$$t = \frac{\ln 2}{0.00002867} \approx 24,180 \text{ yr.}$$



Jack Fields/Science Source

FIGURE 3.1.3 Willard Libby (1908–1980)

CARBON DATING About 1950, a team of scientists at the University of Chicago led by the chemist **Willard Libby** devised a method using a radioactive isotope of carbon as a means of determining the approximate ages of carbonaceous fossilized matter. See Figure 3.1.3. The theory of **carbon dating** is based on the fact that the radioisotope carbon-14 is produced in the atmosphere by the action of cosmic radiation on nitrogen-14. The ratio of the amount of C-14 to the stable C-12 in the atmosphere appears to be a constant, and as a consequence the proportionate amount of the isotope present in all living organisms is the same as that in the atmosphere. When a living organism dies, the absorption of C-14, by breathing, eating, or photosynthesis, ceases. By comparing the proportionate amount of C-14, say, in a fossil with the constant amount ratio found in the atmosphere, it is possible to obtain a reasonable estimation of its age. The method is based on the knowledge of the



FIGURE 3.1.4 A page of the Gnostic Gospel of Judas

Kenneth Garrett/National Geographic Creative

The size and location of the sample caused major difficulties when a team of scientists were invited to use carbon-14 dating on the Shroud of Turin in 1988.



The half-life of uranium-238 is about 4.47 billion years.



half-life of C-14. Libby's calculated value of the half-life of C-14 was approximately 5600 years, and is called the **Libby half-life**. Today the commonly accepted value for the half-life of C-14 is the **Cambridge half-life** that is close to 5730 years. For his work, Libby was awarded the Nobel Prize for chemistry in 1960. Libby's method has been used to date wooden furniture found in Egyptian tombs, the woven flax wrappings of the Dead Sea Scrolls, a recently discovered copy of the Gnostic Gospel of Judas written on papyrus, and the cloth of the enigmatic Shroud of Turin. See Figure 3.1.4 and Problem 12 in Exercises 3.1.

EXAMPLE 3 Age of a Fossil

A fossilized bone is found to contain 0.1% of its original amount of C-14. Determine the age of the fossil.

SOLUTION As in Example 2 the starting point is $A(t) = A_0 e^{kt}$. To determine the value of the decay constant k we use the fact that $\frac{1}{2} A_0 = A(5730)$ or $\frac{1}{2} A_0 = A_0 e^{5730k}$. The last equation implies $5730k = \ln \frac{1}{2} = -\ln 2$ and so we get $k = -(\ln 2)/5730 = -0.00012097$. Therefore $A(t) = A_0 e^{-0.00012097t}$. With $A(t) = 0.001A_0$ we have $0.001A_0 = A_0 e^{-0.00012097t}$ and $-0.00012097t = \ln(0.001) = -\ln 1000$. Thus

$$t = \frac{\ln 1000}{0.00012097} \approx 57,100 \text{ years.}$$

The age found in Example 3 is really at the border of accuracy of this method. The usual carbon-14 technique is limited to about 10 half-lives of the isotope, or roughly 60,000 years. One fundamental reason for this limitation is the relatively short half-life of C-14. There are other problems as well: The chemical analysis needed to obtain an accurate measurement of the remaining C-14 becomes somewhat formidable around the point $0.001A_0$. Moreover, this analysis requires the destruction of a rather large sample of the specimen. If this measurement is accomplished indirectly, based on the actual radioactivity of the specimen, then it is very difficult to distinguish between the radiation from the specimen and the normal background radiation. But recently the use of a particle accelerator has enabled scientists to separate the C-14 from the stable C-12 directly. When the precise value of the ratio of C-14 to C-12 is computed, the accuracy of the dating method can be extended to 70,000–100,000 years. For these reasons and the fact that the C-14 dating is restricted to organic materials, this method is used mainly by archeologists. On the other hand, geologists who are interested in questions about the age of rocks or the age of the Earth use **radiometric-dating** techniques. Radiometric dating, invented by the physicist chemist **Ernest Rutherford** (1871–1937) around 1905, is based on the radioactive decay of a naturally occurring radioactive isotope with a very long half-life and a comparison between a measured quantity of this decaying isotope and one of its decay products using known decay rates. Radiometric methods using potassium-argon, rubidium-strontium, or uranium-lead can give ages of certain kinds of rocks of several billion years. See Problems 5 and 6 in Exercises 3.3 for a brief discussion of the **potassium-argon method of dating**.

NEWTON'S LAW OF COOLING/WARMING In equation (3) of Section 1.3 we saw that the mathematical formulation of Newton's empirical law of cooling/warming of an object is given by the linear first-order differential equation

$$\frac{dT}{dt} = k(T - T_m), \quad (2)$$

where k is a constant of proportionality, $T(t)$ is the temperature of the object for $t > 0$, and T_m is the ambient temperature—that is, the temperature of the medium around the object. In Example 4 we assume that T_m is constant.

EXAMPLE 4 Cooling of a Cake

When a cake is removed from an oven, its temperature is measured at 300°F . Three minutes later its temperature is 200°F . How long will it take for the cake to cool off to a room temperature of 70°F ?

SOLUTION In (2) we make the identification $T_m = 70$. We must then solve the initial-value problem

$$\frac{dT}{dt} = k(T - 70), \quad T(0) = 300 \quad (3)$$

and determine the value of k so that $T(3) = 200$.

Equation (3) is both linear and separable. If we separate variables,

$$\frac{dT}{T - 70} = k dt,$$

yields $\ln |T - 70| = kt + c_1$, and so $T = 70 + c_2 e^{kt}$. When $t = 0$, $T = 300$, so $300 = 70 + c_2$ gives $c_2 = 230$; therefore $T = 70 + 230e^{kt}$. Finally, the measurement $T(3) = 200$ leads to $e^{3k} = \frac{13}{23}$, or $k = \frac{1}{3} \ln \frac{13}{23} = -0.19018$. Thus

$$T(t) = 70 + 230e^{-0.19018t}. \quad (4)$$

We note that (4) furnishes no finite solution to $T(t) = 70$, since $\lim_{t \rightarrow \infty} T(t) = 70$. Yet we intuitively expect the cake to reach room temperature after a reasonably long period of time. How long is “long”? Of course, we should not be disturbed by the fact that the model (3) does not quite live up to our physical intuition. Parts (a) and (b) of Figure 3.1.5 clearly show that the cake will be approximately at room temperature in about one-half hour.

The ambient temperature in (2) need not be a constant but could be a function $T_m(t)$ of time t . See Problem 18 in Exercises 3.1.

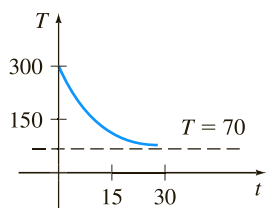
MIXTURES The mixing of two fluids sometimes gives rise to a linear first-order differential equation. When we discussed the mixing of two brine solutions in Section 1.3, we assumed that the rate $A'(t)$ at which the amount of salt in the mixing tank changes was a net rate:

$$\frac{dA}{dt} = (\text{input rate of salt}) - (\text{output rate of salt}) = R_{in} - R_{out}. \quad (5)$$

In Example 5 we solve equation (8) on page 25 of Section 1.3.

EXAMPLE 5 Mixture of Two Salt Solutions

Recall that the large tank considered in Section 1.3 held 300 gallons of a brine solution. Salt was entering and leaving the tank; a brine solution was being pumped into the tank at the rate of 3 gal/min; it mixed with the solution there, and then the mixture was pumped out at the rate of 3 gal/min. The concentration of the salt in the inflow, or solution entering, was 2 lb/gal, so salt was entering the tank at the rate $R_{in} = (2 \text{ lb/gal}) \cdot (3 \text{ gal/min}) = 6 \text{ lb/min}$ and leaving the tank at the rate $R_{out} = (A/300 \text{ lb/gal}) \cdot (3 \text{ gal/min}) = A/100 \text{ lb/min}$. From this data and (5) we get equation (8) of Section 1.3. Let us pose the question: If 50 pounds of salt were dissolved initially in the 300 gallons, how much salt is in the tank after a long time?

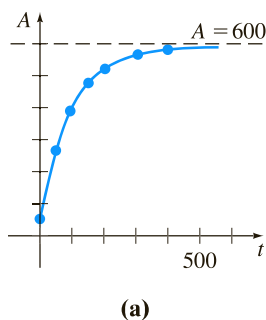


(a)

$T(t)$	t (min)
75°	20.1
74°	21.3
73°	22.8
72°	24.9
71°	28.6
70.5°	32.3

(b)

FIGURE 3.1.5 Temperature of cooling cake in Example 4



t (min)	A (lb)
50	266.41
100	397.67
150	477.27
200	525.57
300	572.62
400	589.93

(b)

FIGURE 3.1.6 Pounds of salt in the tank in Example 5

SOLUTION To find the amount of salt $A(t)$ in the tank at time t , we solve the initial-value problem

$$\frac{dA}{dt} + \frac{1}{100}A = 6, \quad A(0) = 50.$$

Note here that the side condition is the initial amount of salt $A(0) = 50$ in the tank and *not* the initial amount of liquid in the tank. Now since the integrating factor of the linear differential equation is $e^{t/100}$, we can write the equation as

$$\frac{d}{dt}[e^{t/100}A] = 6e^{t/100}.$$

Integrating the last equation and solving for A gives the general solution $A(t) = 600 + ce^{-t/100}$. When $t = 0$, $A = 50$, so we find that $c = -550$. Thus the amount of salt in the tank at time t is given by

$$A(t) = 600 - 550e^{-t/100}. \quad (6)$$

The solution (6) was used to construct the table in Figure 3.1.6(b). Also, it can be seen from (6) and Figure 3.1.6(a) that $A(t) \rightarrow 600$ as $t \rightarrow \infty$. Of course, this is what we would intuitively expect; over a long time the number of pounds of salt in the solution must be $(300 \text{ gal})(2 \text{ lb/gal}) = 600 \text{ lb}$. ■

In Example 5 we assumed that the rate at which the solution was pumped in was the same as the rate at which the solution was pumped out. However, this need not be the case; the mixed brine solution could be pumped out at a rate r_{out} that is faster or slower than the rate r_{in} at which the other brine solution is pumped in. The next example illustrates the case when the mixture is pumped out at rate that is *slower* than the rate at which the brine solution is being pumped into the tank.

EXAMPLE 6 Example 5 Revisited

If the well-stirred solution in Example 5 is pumped out at a slower rate of, say, $r_{out} = 2$ gal/min, then liquid will accumulate in the tank at the rate of

$$r_{in} - r_{out} = (3 - 2) \text{ gal/min} = 1 \text{ gal/min}.$$

After t minutes,

$$(1 \text{ gal/min}) \cdot (t \text{ min}) = t \text{ gal}$$

will accumulate, so the tank will contain $300 + t$ gallons of brine. The concentration of the outflow is then $c(t) = A/(300 + t)$ lb/gal, and the output rate of salt is $R_{out} = c(t) \cdot r_{out}$, or

$$R_{out} = \left(\frac{A}{300 + t} \text{ lb/gal} \right) \cdot (2 \text{ gal/min}) = \frac{2A}{300 + t} \text{ lb/min}.$$

Hence equation (5) becomes

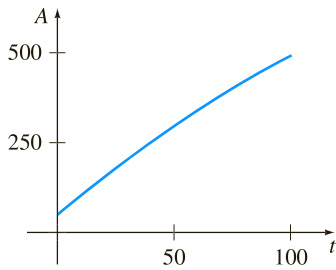
$$\frac{dA}{dt} = 6 - \frac{2A}{300 + t} \quad \text{or} \quad \frac{dA}{dt} + \frac{2}{300 + t}A = 6.$$

The integrating factor for the last equation is

$$e^{\int 2 dt/(300+t)} = e^{2 \ln(300+t)} = e^{\ln(300+t)^2} = (300 + t)^2$$

and so after multiplying by the factor the equation is cast into the form

$$\frac{d}{dt}[(300 + t)^2 A] = 6(300 + t)^2.$$

FIGURE 3.1.7 Graph of $A(t)$ in Example 6

Integrating the last equation gives $(300 + t)^2 A = 2(300 + t)^3 + c$. By applying the initial condition $A(0) = 50$ and solving for A yields the solution $A(t) = 600 + 2t - (4.95 \times 10^7)(300 + t)^{-2}$. As Figure 3.1.7 shows, not unexpectedly, salt builds up in the tank over time, that is, $A \rightarrow \infty$ as $t \rightarrow \infty$. ■

SERIES CIRCUITS For a series circuit containing only a resistor and an inductor, Kirchhoff's second law states that the sum of the voltage drop across the inductor ($L(di/dt)$) and the voltage drop across the resistor (iR) is the same as the impressed voltage ($E(t)$) on the circuit. See Figure 3.1.8.

Thus we obtain the linear differential equation for the current $i(t)$,

$$L \frac{di}{dt} + Ri = E(t), \quad (7)$$

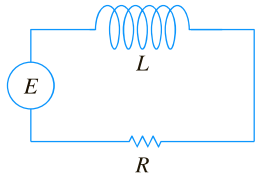
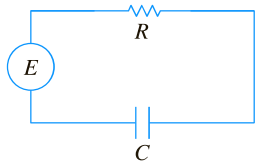
where L and R are known as the inductance and the resistance, respectively. The current $i(t)$ is also called the **response** of the system.

The voltage drop across a capacitor with capacitance C is given by $q(t)/C$, where q is the charge on the capacitor. Hence, for the series circuit shown in Figure 3.1.9, Kirchhoff's second law gives

$$Ri + \frac{1}{C}q = E(t). \quad (8)$$

But current i and charge q are related by $i = dq/dt$, so (8) becomes the linear differential equation

$$R \frac{dq}{dt} + \frac{1}{C}q = E(t). \quad (9)$$

FIGURE 3.1.8 LR -series circuitFIGURE 3.1.9 RC -series circuit

EXAMPLE 7 LR -Series Circuit

A 12-volt battery is connected to a series circuit in which the inductance is $\frac{1}{2}$ henry and the resistance is 10 ohms. Determine the current i if the initial current is zero.

SOLUTION From (7) we see that we must solve the equation

$$\frac{1}{2} \frac{di}{dt} + 10i = 12,$$

subject to $i(0) = 0$. First, we multiply the differential equation by 2 and read off the integrating factor e^{20t} . We then obtain

$$\frac{d}{dt} [e^{20t}i] = 24e^{20t}.$$

Integrating each side of the last equation and solving for i gives $i(t) = \frac{6}{5} + ce^{-20t}$. Now $i(0) = 0$ implies that $0 = \frac{6}{5} + c$ or $c = -\frac{6}{5}$. Therefore the response is $i(t) = \frac{6}{5} - \frac{6}{5}e^{-20t}$. ■

From (4) of Section 2.3 we can write a general solution of (7):

$$i(t) = \frac{e^{-(R/L)t}}{L} \int e^{(R/L)t} E(t) dt + ce^{-(R/L)t}. \quad (10)$$

In particular, when $E(t) = E_0$ is a constant, (10) becomes

$$i(t) = \frac{E_0}{R} + ce^{-(R/L)t}. \quad (11)$$

Note that as $t \rightarrow \infty$, the second term in equation (11) approaches zero. Such a term is usually called a **transient term**; any remaining terms are called the **steady-state** part of the solution. In this case E_0/R is also called the **steady-state current**; for large values of time it appears that the current in the circuit is simply governed by Ohm's law ($E = iR$).

REMARKS

The solution $P(t) = P_0 e^{0.4055t}$ of the initial-value problem in Example 1 described the population of a colony of bacteria at any time $t \geq 0$. Of course, $P(t)$ is a continuous function that takes on *all* real numbers in the interval $P_0 \leq P < \infty$. But since we are talking about a population, common sense dictates that P can take on only positive integer values. Moreover, we would not expect the population to grow continuously—that is, every second, every microsecond, and so on—as predicted by our solution; there may be intervals of time $[t_1, t_2]$ over which there is no growth at all. Perhaps, then, the graph shown in Figure 3.1.10(a) is a more realistic description of P than is the graph of an exponential function. Using a continuous function to describe a discrete phenomenon is often more a matter of convenience than of accuracy. However, for some purposes we may be satisfied if our model describes the system fairly closely when viewed macroscopically in time, as in Figures 3.1.10(b) and 3.1.10(c), rather than microscopically, as in Figure 3.1.10(a).

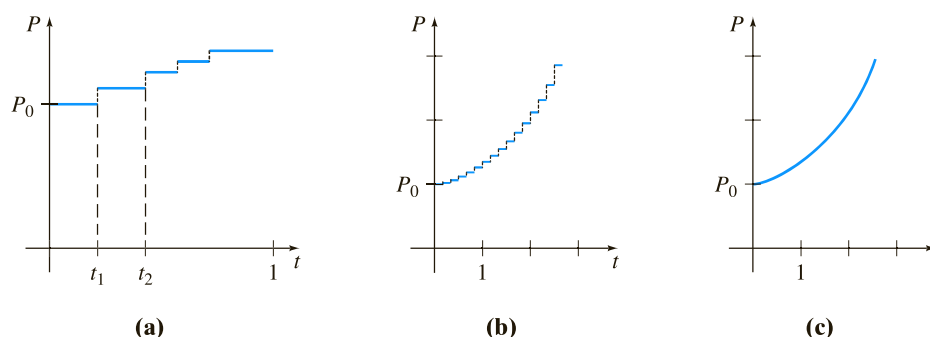


FIGURE 3.1.10 Population growth is a discrete process

EXERCISES 3.1

Answers to selected odd-numbered problems begin on page ANS-3.

Growth and Decay

- The population of a community is known to increase at a rate proportional to the number of people present at time t . If an initial population P_0 has doubled in 5 years, how long will it take to triple? To quadruple?
- Suppose it is known that the population of the community in Problem 1 is 10,000 after 3 years. What was the initial population P_0 ? What will be the population in 10 years? How fast is the population growing at $t = 10$?
- The population of a town grows at a rate proportional to the population present at time t . The initial population of 500 increases by 15% in 10 years. What will be the population in 30 years? How fast is the population growing at $t = 30$?
- The population of bacteria in a culture grows at a rate proportional to the number of bacteria present at time t . After 3 hours it is observed that 400 bacteria are present. After 10 hours 2000 bacteria are present. What was the initial number of bacteria?
- The radioactive isotope of lead, Pb-209, decays at a rate proportional to the amount present at time t and has a half-life of 3.3 hours. If 1 gram of this isotope is present initially, how long will it take for 90% of the lead to decay?
- Initially 100 milligrams of a radioactive substance was present. After 6 hours the mass had decreased by 3%. If the rate of decay is proportional to the amount of the substance present at time t , find the amount remaining after 24 hours.

7. Determine the half-life of the radioactive substance described in Problem 6.
8. (a) Consider the initial-value problem $dA/dt = kA$, $A(0) = A_0$ as the model for the decay of a radioactive substance. Show that, in general, the half-life T of the substance is $T = -(\ln 2)/k$.
 (b) Show that the solution of the initial-value problem in part (a) can be written $A(t) = A_0 2^{-t/T}$.
 (c) If a radioactive substance has the half-life T given in part (a), how long will it take an initial amount A_0 of the substance to decay to $\frac{1}{8}A_0$?
9. When a vertical beam of light passes through a transparent medium, the rate at which its intensity I decreases is proportional to $I(t)$, where t represents the thickness of the medium (in feet). In clear seawater, the intensity 3 feet below the surface is 25% of the initial intensity I_0 of the incident beam. What is the intensity of the beam 15 feet below the surface?
10. When interest is compounded continuously, the amount of money increases at a rate proportional to the amount S present at time t , that is, $dS/dt = rS$, where r is the annual rate of interest.
 (a) Find the amount of money accrued at the end of 5 years when \$5000 is deposited in a savings account drawing $5\frac{3}{4}\%$ annual interest compounded continuously.
 (b) In how many years will the initial sum deposited have doubled?
 (c) Use a calculator to compare the amount obtained in part (a) with the amount $S = 5000(1 + \frac{1}{4}(0.0575))^{5(4)}$ that is accrued when interest is compounded quarterly.

Carbon Dating

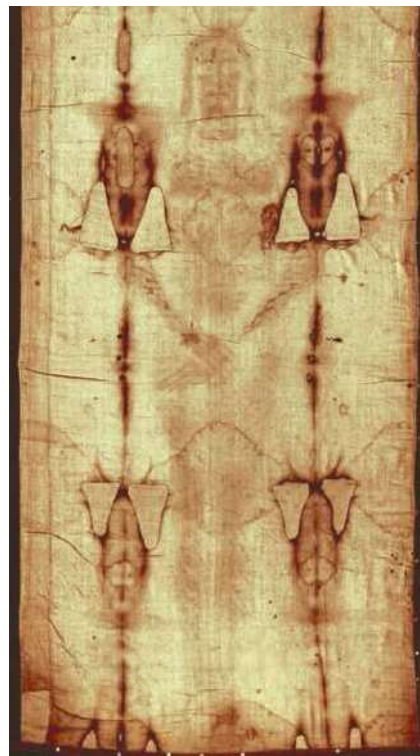
11. Archaeologists used pieces of burned wood, or charcoal, found at the site to date prehistoric paintings and drawings on walls and ceilings of a cave in Lascaux, France. See Figure 3.1.11. Use the information on page 87 to determine the approximate age of a piece of burned wood, if it was found that 85.5% of the C-14 found in living trees of the same type had decayed.



Rock painting showing a horse and a cow, c. 17,000 bc (cave painting). Prehistoric/Caves of Lascaux, Dordogne, France/Bridgeman Images

FIGURE 3.1.11 Cave wall painting in Problem 11

12. The Shroud of Turin, which shows the negative image of the body of a man who appears to have been crucified, is believed by many to be the burial shroud of Jesus of Nazareth. See Figure 3.1.12. In 1988 the Vatican granted permission to have the shroud carbon-dated. Three independent scientific laboratories analyzed the cloth and concluded that the shroud was approximately 660 years old, an age consistent with its historical appearance. Using this age, determine what percentage of the original amount of C-14 remained in the cloth as of 1988.



Source: Wikipedia.org

FIGURE 3.1.12 Shroud image in Problem 12

Newton's Law of Cooling/Warming

13. A thermometer is removed from a room where the temperature is 70°F and is taken outside, where the air temperature is 10°F . After one-half minute the thermometer reads 50°F . What is the reading of the thermometer at $t = 1$ min? How long will it take for the thermometer to reach 15°F ?
14. A thermometer is taken from an inside room to the outside, where the air temperature is 5°F . After 1 minute the thermometer reads 55°F , and after 5 minutes it reads 30°F . What is the initial temperature of the inside room?
15. A small metal bar, whose initial temperature was 20°C , is dropped into a large container of boiling water. How long will it take the bar to reach 90°C if it is known that its temperature increases 2° in 1 second? How long will it take the bar to reach 98°C ?
16. Two large containers A and B of the same size are filled with different fluids. The fluids in containers A and B are maintained at 0°C and 100°C , respectively. A small metal bar, whose initial temperature is 100°C , is lowered into container A . After 1 minute the temperature of the bar is 90°C . After 2 minutes the bar is removed and instantly transferred to the other container. After 1 minute in container B the temperature of the bar rises 10° . How long, measured from the start of the entire process, will it take the bar to reach 99.9°C ?
17. A thermometer reading 70°F is placed in an oven preheated to a constant temperature. Through a glass window in the oven door, an observer records that the thermometer reads 110°F after $\frac{1}{2}$ minute and 145°F after 1 minute. How hot is the oven?
18. At $t = 0$ a sealed test tube containing a chemical is immersed in a liquid bath. The initial temperature of the chemical in the test tube is 80°F . The liquid bath has a controlled

temperature (measured in degrees Fahrenheit) given by $T_m(t) = 100 - 40e^{-0.1t}$, $t \geq 0$, where t is measured in minutes.

- (a) Assume that $k = -0.1$ in (2). Before solving the IVP, describe in words what you expect the temperature $T(t)$ of the chemical to be like in the short term. In the long term.
 - (b) Solve the initial-value problem. Use a graphing utility to plot the graph of $T(t)$ on time intervals of various lengths. Do the graphs agree with your predictions in part (a)?
19. A dead body was found within a closed room of a house where the temperature was a constant 70°F . At the time of discovery the core temperature of the body was determined to be 85°F . One hour later a second measurement showed that the core temperature of the body was 80°F . Assume that the time of death corresponds to $t = 0$ and that the core temperature at that time was 98.6°F . Determine how many hours elapsed before the body was found. [Hint: Let $t_1 > 0$ denote the time that the body was discovered.]
20. The rate at which a body cools also depends on its exposed surface area S . If S is a constant, then a modification of (2) is

$$\frac{dT}{dt} = kS(T - T_m),$$

where $k < 0$ and T_m is a constant. Suppose that two cups A and B are filled with coffee at the same time. Initially, the temperature of the coffee is 150°F . The exposed surface area of the coffee in cup B is twice the surface area of the coffee in cup A . After 30 min the temperature of the coffee in cup A is 100°F . If $T_m = 70^\circ\text{F}$, then what is the temperature of the coffee in cup B after 30 min?

Mixtures

21. A tank contains 200 liters of fluid in which 30 grams of salt is dissolved. Brine containing 1 gram of salt per liter is then pumped into the tank at a rate of 4 L/min; the well-mixed solution is pumped out at the same rate. Find the number $A(t)$ of grams of salt in the tank at time t .
22. Solve Problem 21 assuming that pure water is pumped into the tank.
23. A large tank is filled to capacity with 500 gallons of pure water. Brine containing 2 pounds of salt per gallon is pumped into the tank at a rate of 5 gal/min. The well-mixed solution is pumped out at the same rate. Find the number $A(t)$ of pounds of salt in the tank at time t .
24. In Problem 23, what is the concentration $c(t)$ of the salt in the tank at time t ? At $t = 5$ min? What is the concentration of the salt in the tank after a long time, that is, as $t \rightarrow \infty$? At what time is the concentration of the salt in the tank equal to one-half this limiting value?
25. Solve Problem 23 under the assumption that the solution is pumped out at a faster rate of 10 gal/min. When is the tank empty?
26. Determine the amount of salt in the tank at time t in Example 5 if the concentration of salt in the inflow is variable and given by $c_{in}(t) = 2 + \sin(t/4)$ lb/gal. Without actually graphing, conjecture what the solution curve of the IVP should look like. Then use a graphing utility to plot the graph of the solution on the interval $[0, 300]$. Repeat for the interval $[0, 600]$ and compare your graph with that in Figure 3.1.6(a).
27. A large tank is partially filled with 100 gallons of fluid in which 10 pounds of salt is dissolved. Brine containing $\frac{1}{2}$ pound of salt per gallon is pumped into the tank at a rate of 6 gal/min. The well-mixed solution is then pumped out at a slower rate of 4 gal/min. Find the number of pounds of salt in the tank after 30 minutes.
28. In Example 5 the size of the tank containing the salt mixture was not given. Suppose, as in the discussion following Example 5, that the rate at which brine is pumped into the tank is 3 gal/min but that the well-stirred solution is pumped out at a rate of 2 gal/min. It stands to reason that since brine is accumulating in the tank at the rate of 1 gal/min, any finite tank must eventually overflow. Now suppose that the tank has an open top and has a total capacity of 400 gallons.
- (a) When will the tank overflow?
 - (b) What will be the number of pounds of salt in the tank at the instant it overflows?
 - (c) Assume that although the tank is overflowing, brine solution continues to be pumped in at a rate of 3 gal/min and the well-stirred solution continues to be pumped out at a rate of 2 gal/min. Devise a method for determining the number of pounds of salt in the tank at $t = 150$ minutes.
 - (d) Determine the number of pounds of salt in the tank as $t \rightarrow \infty$. Does your answer agree with your intuition?
 - (e) Use a graphing utility to plot the graph of $A(t)$ on the interval $[0, 500]$.

Series Circuits

29. A 30-volt electromotive force is applied to an LR -series circuit in which the inductance is 0.1 henry and the resistance is 50 ohms. Find the current $i(t)$ if $i(0) = 0$. Determine the current as $t \rightarrow \infty$.
30. Solve equation (7) under the assumption that $E(t) = E_0 \sin \omega t$ and $i(0) = i_0$.
31. A 100-volt electromotive force is applied to an RC -series circuit in which the resistance is 200 ohms and the capacitance is 10^{-4} farad. Find the charge $q(t)$ on the capacitor if $q(0) = 0$. Find the current $i(t)$.
32. A 200-volt electromotive force is applied to an RC -series circuit in which the resistance is 1000 ohms and the capacitance is 5×10^{-6} farad. Find the charge $q(t)$ on the capacitor if $i(0) = 0.4$. Determine the charge and current at $t = 0.005$ s. Determine the charge as $t \rightarrow \infty$.
33. An electromotive force

$$E(t) = \begin{cases} 120, & 0 \leq t \leq 20 \\ 0, & t > 20 \end{cases}$$

is applied to an LR -series circuit in which the inductance is 20 henries and the resistance is 2 ohms. Find the current $i(t)$ if $i(0) = 0$.

34. An LR -series circuit has a variable inductor with the inductance defined by

$$L(t) = \begin{cases} 1 - \frac{1}{10}t, & 0 \leq t < 10 \\ 0, & t > 10. \end{cases}$$

Find the current $i(t)$ if the resistance is 0.2 ohm, the impressed voltage is $E(t) = 4$, and $i(0) = 0$. Graph $i(t)$.

Additional Linear Models

35. Air Resistance In (14) of Section 1.3 we saw that a differential equation describing the velocity v of a falling mass subject to air resistance proportional to the instantaneous velocity is

$$m \frac{dv}{dt} = mg - kv,$$

where $k > 0$ is a constant of proportionality. The positive direction is downward.

- Solve the equation subject to the initial condition $v(0) = v_0$.
 - Use the solution in part (a) to determine the limiting, or terminal, velocity of the mass. We saw how to determine the terminal velocity without solving the DE in Problem 40 in Exercises 2.1.
 - If the distance s , measured from the point where the mass was released above ground, is related to velocity v by $ds/dt = v(t)$, find an explicit expression for $s(t)$ if $s(0) = 0$.
- 36. How High?—No Air Resistance** Suppose a small cannonball weighing 16 pounds is shot vertically upward, as shown in Figure 3.1.13, with an initial velocity $v_0 = 300$ ft/s. The answer to the question “How high does the cannonball go?” depends on whether we take air resistance into account.
- Suppose air resistance is ignored. If the positive direction is upward, then a model for the state of the cannonball is given by $d^2s/dt^2 = -g$ (equation (12) of Section 1.3). Since $ds/dt = v(t)$ the last differential equation is the same as $dv/dt = -g$, where we take $g = 32$ ft/s². Find the velocity $v(t)$ of the cannonball at time t .
 - Use the result obtained in part (a) to determine the height $s(t)$ of the cannonball measured from ground level. Find the maximum height attained by the cannonball.

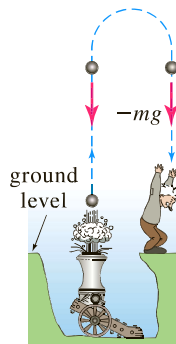


FIGURE 3.1.13 Find the maximum height of the cannonball in Problem 36

- 37. How High?—Linear Air Resistance** Repeat Problem 36, but this time assume that air resistance is proportional to instantaneous velocity. It stands to reason that the maximum height attained by the cannonball must be *less* than that in part (b) of Problem 36. Show this by supposing that the constant of proportionality is $k = 0.0025$. [Hint: Slightly modify the differential equation in Problem 35.]
- 38. Skydiving** A skydiver weighs 125 pounds, and her parachute and equipment combined weigh another 35 pounds. After

exiting from a plane at an altitude of 15,000 feet, she waits 15 seconds and opens her parachute. Assume that the constant of proportionality in the model in Problem 35 has the value $k = 0.5$ during free fall and $k = 10$ after the parachute is opened. Assume that her initial velocity on leaving the plane is zero. What is her velocity and how far has she traveled 20 seconds after leaving the plane? See Figure 3.1.14. How does her velocity at 20 seconds compare with her terminal velocity? How long does it take her to reach the ground? [Hint: Think in terms of two distinct IVPs.]

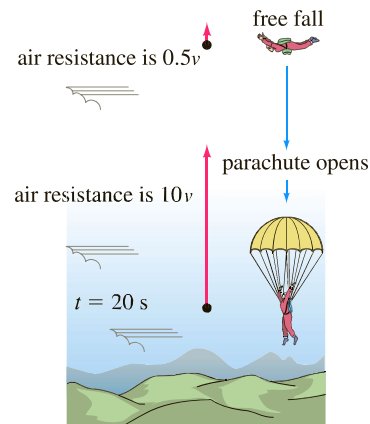


FIGURE 3.1.14 Find the time to reach the ground in Problem 38

- 39. Rocket Motion** Suppose a small single-stage rocket of total mass $m(t)$ is launched vertically, the positive direction is upward, the air resistance is linear, and the rocket consumes its fuel at a constant rate. In Problem 22 of Exercises 1.3 you were asked to use Newton's second law of motion in the form given in (17) of that exercise set to show that a mathematical model for the velocity $v(t)$ of the rocket is given by

$$\frac{dv}{dt} + \frac{k - \lambda}{m_0 - \lambda t} v = -g + \frac{R}{m_0 - \lambda t},$$

where k is the air resistance constant of proportionality, λ is the constant rate at which fuel is consumed, R is the thrust of the rocket, $m(t) = m_0 - \lambda t$, m_0 is the total mass of the rocket at $t = 0$, and g is the acceleration due to gravity.

- Find the velocity $v(t)$ of the rocket if $m_0 = 200$ kg, $R = 2000$ N, $\lambda = 1$ kg/s, $g = 9.8$ m/s², $k = 3$ kg/s, and $v(0) = 0$.
 - Use $ds/dt = v$ and the result in part (a) to find the height $s(t)$ of the rocket at time t .
- 40. Rocket Motion—Continued** In Problem 39 suppose of the rocket's initial mass m_0 that 50 kg is the mass of the fuel.
- What is the burnout time t_b , or the time at which all the fuel is consumed?
 - What is the velocity of the rocket at burnout?
 - What is the height of the rocket at burnout?
 - Why would you expect the rocket to attain an altitude higher than the number in part (b)?
 - After burnout what is a mathematical model for the velocity of the rocket?

- 41. Evaporating Raindrop** As a raindrop falls, it evaporates while retaining its spherical shape. If we make the further assumptions that the rate at which the raindrop evaporates is proportional to its surface area and that air resistance is negligible, then a model for the velocity $v(t)$ of the raindrop is

$$\frac{dv}{dt} + \frac{3(k/\rho)}{(k/\rho)t + r_0}v = g.$$

Here ρ is the density of water, r_0 is the radius of the raindrop at $t = 0$, $k < 0$ is the constant of proportionality, and the downward direction is taken to be the positive direction.

- (a) Solve for $v(t)$ if the raindrop falls from rest.
- (b) Reread Problem 36 of Exercises 1.3 and then show that the radius of the raindrop at time t is $r(t) = (k/\rho)t + r_0$.
- (c) If $r_0 = 0.01$ ft and $r = 0.007$ ft 10 seconds after the raindrop falls from a cloud, determine the time at which the raindrop has evaporated completely.
- 42. Fluctuating Population** The differential equation $dP/dt = (k \cos t)P$, where k is a positive constant, is a mathematical model for a population $P(t)$ that undergoes yearly seasonal fluctuations. Solve the equation subject to $P(0) = P_0$. Use a graphing utility to graph the solution for different choices of P_0 .

- 43. Population Model** In one model of the changing population $P(t)$ of a community, it is assumed that

$$\frac{dP}{dt} = \frac{dB}{dt} - \frac{dD}{dt},$$

where dB/dt and dD/dt are the birth and death rates, respectively.

- (a) Solve for $P(t)$ if $dB/dt = k_1P$ and $dD/dt = k_2P$.
- (b) Analyze the cases $k_1 > k_2$, $k_1 = k_2$, and $k_1 < k_2$.
- 44. Constant-Harvest Model** A model that describes the population of a fishery in which harvesting takes place at a constant rate is given by

$$\frac{dP}{dt} = kP - h,$$

where k and h are positive constants.

- (a) Solve the DE subject to $P(0) = P_0$.
- (b) Describe the behavior of the population $P(t)$ for increasing time in the three cases $P_0 > h/k$, $P_0 = h/k$, and $0 < P_0 < h/k$.
- (c) Use the results from part (b) to determine whether the fish population will ever go extinct in finite time, that is, whether there exists a time $T > 0$ such that $P(T) = 0$. If the population goes extinct, then find T .
- 45. Drug Dissemination** A mathematical model for the rate at which a drug disseminates into the bloodstream is given by

$$\frac{dx}{dt} = r - kx,$$

where r and k are positive constants. The function $x(t)$ describes the concentration of the drug in the bloodstream at time t .

- (a) Since the DE is autonomous, use the phase portrait concept of Section 2.1 to find the limiting value of $x(t)$ as $t \rightarrow \infty$.
- (b) Solve the DE subject to $x(0) = 0$. Sketch the graph of $x(t)$ and verify your prediction in part (a). At what time is the concentration one-half this limiting value?
- 46. Memorization** When forgetfulness is taken into account, the rate of memorization of a subject is given by

$$\frac{dA}{dt} = k_1(M - A) - k_2A,$$

where $k_1 > 0$, $k_2 > 0$, $A(t)$ is the amount memorized in time t , M is the total amount to be memorized, and $M - A$ is the amount remaining to be memorized.

- (a) Since the DE is autonomous, use the phase portrait concept of Section 2.1 to find the limiting value of $A(t)$ as $t \rightarrow \infty$. Interpret the result.
- (b) Solve the DE subject to $A(0) = 0$. Sketch the graph of $A(t)$ and verify your prediction in part (a).

- 47. Heart Pacemaker** A heart pacemaker, shown in Figure 3.1.15, consists of a switch, a battery, a capacitor, and the heart as a resistor. When the switch S is at P , the capacitor charges; when S is at Q , the capacitor discharges, sending an electrical stimulus to the heart. In Problem 58 in Exercises 2.3 we saw that during this time the electrical stimulus is being applied to the heart, the voltage E across the heart satisfies the linear DE

$$\frac{dE}{dt} = -\frac{1}{RC}E.$$

- (a) Let us assume that over the time interval of length t_1 , $0 < t < t_1$, the switch S is at position P shown in Figure 3.1.15 and the capacitor is being charged. When the switch is moved to position Q at time t_1 the capacitor discharges, sending an impulse to the heart over the time interval of length t_2 : $t_1 \leq t < t_1 + t_2$. Thus over the initial charging/discharging interval $0 < t < t_1 + t_2$ the voltage to the heart is actually modeled by the piecewise-linear differential equation

$$\frac{dE}{dt} = \begin{cases} 0, & 0 \leq t < t_1 \\ -\frac{1}{RC}E, & t_1 \leq t < t_1 + t_2. \end{cases}$$

By moving S between P and Q , the charging and discharging over time intervals of lengths t_1 and t_2 is

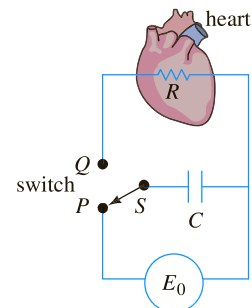


FIGURE 3.1.15 Model of a pacemaker in Problem 47

repeated indefinitely. Suppose $t_1 = 4$ s, $t_2 = 2$ s, $E_0 = 12$ V, and $E(0) = 0$, $E(4) = 12$, $E(6) = 0$, $E(10) = 12$, $E(12) = 0$, and so on. Solve for $E(t)$ for $0 \leq t \leq 24$.

- (b) Suppose for the sake of illustration that $R = C = 1$. Use a graphing utility to graph the solution for the IVP in part (a) for $0 \leq t \leq 24$.

48. Sliding Box (a) A box of mass m slides down an inclined plane that makes an angle θ with the horizontal as shown in Figure 3.1.16. Find a differential equation for the velocity $v(t)$ of the box at time t in each of the following three cases:

- (i) No sliding friction and no air resistance
- (ii) With sliding friction and no air resistance
- (iii) With sliding friction and air resistance

In cases (ii) and (iii), use the fact that the force of friction opposing the motion of the box is μN , where μ is the coefficient of sliding friction and N is the normal component of the weight of the box. In case (iii) assume that air resistance is proportional to the instantaneous velocity.

- (b) In part (a), suppose that the box weighs 96 pounds, that the angle of inclination of the plane is $\theta = 30^\circ$, that the coefficient of sliding friction is $\mu = \sqrt{3}/4$, and that the additional retarding force due to air resistance is numerically equal to $\frac{1}{4}v$. Solve the differential equation in each of the three cases, assuming that the box starts from rest from the highest point 50 ft above ground.

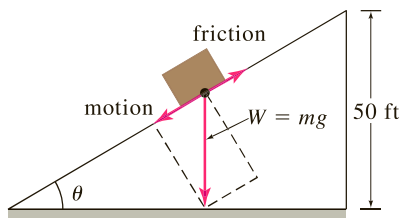


FIGURE 3.1.16 Box sliding down inclined plane in Problem 48

49. Sliding Box—Continued (a) In Problem 48 let $s(t)$ be the distance measured down the inclined plane from the highest point. Use $ds/dt = v(t)$ and the solution for each of the three cases in part (b) of Problem 48 to find the time that it takes the box to slide completely down the inclined plane. A root-finding application of a CAS may be useful here.

- (b) In the case in which there is friction ($\mu \neq 0$) but no air resistance, explain why the box will not slide down the plane starting from rest from the highest point above ground when the inclination angle θ satisfies $\tan \theta \leq \mu$.
- (c) The box will slide downward on the plane when $\tan \theta \leq \mu$ if it is given an initial velocity $v(0) = v_0 > 0$. Suppose that $\mu = \sqrt{3}/4$ and $\theta = 23^\circ$. Verify that $\tan \theta \leq \mu$. How far will the box slide down the plane if $v_0 = 1$ ft/s?
- (d) Using the values $\mu = \sqrt{3}/4$ and $\theta = 23^\circ$, approximate the smallest initial velocity v_0 that can be given to the box so that, starting at the highest point 50 ft above ground, it will slide completely down the inclined plane. Then find the corresponding time it takes to slide down the plane.

50. What Goes Up . . . (a) It is well known that the model in which air resistance is ignored, part (a) of Problem 36, predicts that the time t_d it takes the cannonball to attain its maximum height is the same as the time t_d it takes the cannonball to fall from the maximum height to the ground. Moreover, the magnitude of the impact velocity v_i will be the same as the initial velocity v_0 of the cannonball. Verify both of these results.

- (b) Then, using the model in Problem 37 that takes air resistance into account, compare the value of t_a with t_d and the value of the magnitude of v_i with v_0 . A root-finding application of a CAS (or graphic calculator) may be useful here.

3.2 Nonlinear Models

INTRODUCTION We finish our study of single first-order differential equations with an examination of some nonlinear models.

POPULATION DYNAMICS If $P(t)$ denotes the size of a population at time t , the model for exponential growth begins with the assumption that $dP/dt = kP$ for some $k > 0$. In this model, the **relative**, or **specific**, **growth rate** defined by

$$\frac{dP/dt}{P} \quad (1)$$

is a constant k . True cases of exponential growth over long periods of time are hard to find because the limited resources of the environment will at some time exert restrictions on the growth of a population. Thus for other models, (1) can be expected to decrease as the population P increases in size.